(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

__III_____Semester, November, 2022

Paper Code: EEC-211_____ Subject: ___Electrical Machines-I_____

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	What are the advantages of analysing energy conversion by field energy methods?	1	1
2	Explain Iron Losses?	2	1
3	What are the different types of torque in machines?	2	1
4	What is commutation and method to improve commutation?	2.5	2
5	Explain armature reaction and what are the disadvantages of armature reaction?	2.5	2

Q2 (a) What do you mean by electromechanical energy conversion system, draw flow diagram for Mechanical to Electrical Energy Conversion. (5)

Q2 (b) What is the relation for field energy and field coenergy relation for linear case? (5)

Q3 (a) Explain Critical Field resistance and Critical speed? (5)

Q3 (b) What are the conditions for voltage build up in the Shunt Generator? (5)

Q4 (a) What are the different methods to control speed of DC shunt and DC Series motor? (5)

Q4 (b) A 400 V DC shunt machines takes 7 A at No load, when operates as a motor. The Armature resistance and field resistances are 0.25Ω and 200 Ω, respectively. Calculate efficiency of machine: (5)

When run as a motor taking a line current of 70 A at 400V.

When run as a Generator delivering a line current of 70 A at 400 V.

The stray load loss is 1.1% of the output.